Day 3: Switch and Router Configuration BasicsTheory

Task 1 : Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

To create two VLANs named Sales and Support on a Cisco switch using Packet Tracer.

Commands Used

enable

configure terminal

vlan 10

name Sales

exit

vlan 20

name Support

exit

end

show vlan brief

Result

VLAN 10 named Sales and VLAN 20 named Support were created on the switch. The show vlan brief command was used to verify their names and status.

Task 2 : Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.

To assign three switch ports to VLAN 10 (Sales) and two switch ports to VLAN 20 (Support).

Commands Used

enable

configure terminal

interface range fastEthernet 0/1-3

switchport mode access

switchport access vlan 10

exit

interface range fastEthernet 0/4-5

switchport mode access

switchport access vlan 20

exit

end

show vlan brief

Verification

The show vlan brief command was used to verify the VLAN membership.

- VLAN 10 (Sales): Fa0/1, Fa0/2, Fa0/3
- VLAN 20 (Support): Fa0/4, Fa0/5

Result

Three ports were assigned to Sales and two ports were assigned to Support.

Task 3 : Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

To configure a static route between two routers so that devices on different LANs can communicate.

IP Addressing

- LAN 1: 192.168.1.0/24
- LAN 2: 192.168.2.0/24
- Router link: 10.0.0.0/30

Static Route on Router 1

ip route 192.168.2.0 255.255.255.0 10.0.0.2

This route forwards packets for LAN 2 to Router 2.

Return Route on Router 2

ip route 192.168.1.0 255.255.255.0 10.0.0.1

This route allows reply packets to reach LAN 1.

Verification

The show ip route command was used to check the routing table. A ping test from PC0 to PC1 using ping 192.168.2.10 can verify end-to-end connectivity.

Task 4 : Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.

To test communication between devices in different VLANs and understand the need for inter-VLAN routing.

Procedure

1. Created VLAN 10 named Sales and VLAN 20 named Support.
2. Assigned PC0 to VLAN 10 and PC1 to VLAN 20.
3. Configured PC0 with IP address 192.168.10.10/24 and PC1 with 192.168.20.10/24.
4. From PC0, executed ping 192.168.20.10.

Observation

The ping is expected to fail because the PCs belong to different VLANs and IP subnets. A Layer 2 switch does not route packets between VLANs by default.

Solution

Inter-VLAN communication can be enabled by configuring Router-on-a-Stick with subinterfaces and a trunk link, or by using a Layer 3 switch with appropriate VLAN interfaces and routing.